Models of Equity Prices

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We will frame our discussion around the equilibrium valuation of equity assets, but as we've seen in previous discussions, the methods we develop will have broader applications to any asset class that entitles its owner to a flow of random future payoffs. In the context of equity, we refer to these payoffs as *dividends*. Unlike our discussion of bonds, however, equity assets can be thought of as *perpetuities* since they have no fixed maturity or expiration date.[1]

[1] We needn't take this too literally, since we know that in the future firms can cease to exist due to bankruptcy, mergers, *etc*. The main point is that these are not pre-scheduled events like the maturity of a bond or the expiration of an option. Rather they are just unpredictable random future outcomes that will affect the probability distribution of future payoffs. By the way, we do have a few examples of very long-lived equity assets. The oldest stock traded in the US is BNY Mellon. The Bank of New York was founded by Alexander Hamilton in 1784 and began trading on the New York Stock Exchange in 1792. It merged with Mellon Bank – another senior citizen of the equity world founded in 1869 – in 2007 and continues to be a profitable venture with little risk of shutting down in the near future.

The state of the economy is given by the stochastic process $\{x_t\}$ which we assume to be a well-behaved process that satisfies the Markov condition, *i.e.*, all date-t information that affects the probability distribution of x_{t+1} is summarized by the value of x_t. Consider the price at date t, denoted $P(x_t)$, of an ownership claim to a perpetual sequence of random future dividends, $\{D(x_{t+1}), D(x_{t+2}), D(x_{t+3}), \ldots\}$. If the pricing kernel is given by $m(x_{t+1}, x_t)$, then we can write the price as

$$P(x_t) \quad = \quad E\big[m(x_{t+1}, x_t)\underbrace{[P(x_{t+1}) + D(x_{t+1})]}_{\text{date-}t\text{ payoff}} \mid x_t\big].$$

In other words, at date $t+1$ when the state of the economy is x_{t+1}, the owner of this assets collects both a dividend, $D(x_{t+1})$, and continues to own the asset whose value is given by $P(x_{t+1})$.

Note that $P(x_t)$ is known at date t, so we can divide both sides of this equation by $P(x_t)$, and since $P(x_t)$ is just a constant number at date t, it will pass through the conditional expected value operation which results in an equivalent representation in terms of the asset's return

$$1 \quad = \quad E\big[m(x_{t+1}, x_t)R(x_{t+1}, x_t) \mid x_t\big],$$

where

$$R(x_{t+1}, x_t) \quad = \quad \frac{P(x_{t+1})}{P(x_t)} + \frac{D(x_{t+1})}{P(x_t)}.$$

The first term in this return is often referred to as the *capital gain* and the second term as the *dividend yield*.

We can also *iterate forward* the recursive equation for the price and apply the law of iterated expectations to obtain a familiar *present*

discounted value form for the price

$$P(x_t) = \sum_{\tau=1}^{\infty} E\Big[\underbrace{\Big(\prod_{j=1}^{\tau} m(x_{t+j}, x_{t+j-1})\Big)}_{t+\tau \text{ to } t \text{ pricing kernel}} D(x_{t+\tau}) \;\Big|\; x_t\Big].$$

We often encounter situations in which the dividend process is not stationary, but rather is growing steadily over time. In this case the price will also grow steadily over time. In other words, neither the dividend or the price will be a well-behaved stochastic process. In most instances, however, the growth in the dividend will be well-behaved so the growth rate becomes the stochastic process that we model. A simple rescaling of the asset pricing equation returns the model to a more useful form. That is, divide both sides of the asset pricing equation by $D(x_t)$. Again, since $D(x_t)$ is known at date t it will pass through the conditional expected value operation and result in the equation

$$\begin{aligned}
\frac{P(x_t)}{D(x_t)} &= E\Big[m(x_{t+1}, x_t)\Big[\frac{P(x_{t+1})}{D(x_t)} + \frac{D(x_{t+1})}{D(x_t)}\Big] \;\Big|\; x_t\Big] \\
&= E\Big[m(x_{t+1}, x_t)\Big[\frac{P(x_{t+1})}{D(x_{t+1})}\frac{D(x_{t+1})}{D(x_t)} + \frac{D(x_{t+1})}{D(x_t)}\Big] \;\Big|\; x_t\Big] \\
\Rightarrow p(x_t) &= E\big[m(x_{t+1}, x_t) d(x_{t+1})[p(x_{t+1}) + 1] \;\big|\; x_t\big],
\end{aligned}$$

where we have implicitly defined dividend growth, $d(x_{t+1}) = D(x_{t+1})/D(x_t)$, and the *price-dividend ratio*, $p(x_t) = P(x_t)/D(x_t)$.

Example: The Gordon growth model. Assume that investors are risk neutral so that the pricing kernel for all dates and for all states is a constant, $m(x_{t+1}, x_t) = 1/(1+r)$, where r is the one-period interest rate. Further, assume that the state variable is an *iid* stochastic process. This implies that the dividend growth rate, $d(x_{t+1})$ is also *iid*. Denote its mean as $E[d(x_{t+1})] = 1 + g$. In addition, assume that $r > g$, which is required to keep the infinite sum of expected discounted future dividends finite.[2] Note that in general, the price-dividend ratio, $p(x_t)$, depends on x_t because both the pricing kernel could potentially depend on x_t, and because of the dependence of the conditional expectation on the value of x_t. In this example, however, the pricing kernel is a constant and does not depend on x_t. Moreover, x_t is assumed to be an *iid* stochastic process so the conditional distribution of x_{t+1} does not depend on x_t. Therefore, $p(x_t)$ will be a constant at all time periods and for all values of the state. In other words, we can

[2] The expected dividend at date $t+\tau$ is $D_t(1+g)^\tau$. The pricing kernel between $t+\tau$ and t is $1/(1+r)^\tau$. Therefore the date-t value of $D_{t+\tau}$ is $D_t[(1+g)/(1+r)]^\tau$. To insure that this converges to 0 as $\tau \to \infty$, we need $(1+g)/(1+r) < 1$ or simply $r > g$.

write the equation for the price-dividend ratio as

$$\begin{aligned} p &= \frac{E[d(x_{t+1})](p+1)}{1+r} \\ \Rightarrow p &= \frac{1+g}{r-g}\,, \end{aligned}$$

which implies that the price (rather than the price-dividend ratio) is given by

$$P = \frac{E(D_{t+1})}{r-g}\,.$$

Clearly, this is a highly-stylized example.We rarely encounter situations in which the dynamics of the state of the economy are well-approximated by an *iid* process. And we rarely encounter situations in which the representative agent is well-approximated by a risk-neutral investor. Nonetheless, this simple formula is often useful for understanding the scale of prices relative to dividends. That is, the expected dividend is 1, $r = 0.02$, and $g = 0.01$, then this simple model predicts that the price will be two orders of magnitude bigger, $P = 1/0.01 = 100$. Moreover, it forces us to think about relationship between perpetual growth g of a cashflow versus the perpetual discounting of that cashflow at a rate of r.

Example: A simple AR(1) model. We build the dynamic model using our customary ingredients:

- State variable dynamics:

  $$x_t = \varphi x_{t-1} + \sigma w_t,$$

  where $-1 < \varphi < 1$ and $\sigma > 0$ are constant parameters, and w_t is *iid* $\mathcal{N}(0,1)$.

- Pricing kernel:

  $$\log m_{t+1} = \mu + x_t + \lambda w_{t+1},$$

  where μ and λ are constant parameters.

- Dividend growth:

  $$\log d(x_t) = \delta_0 + \delta_1 x_t,$$

  where δ_0 and δ_1 are constant paramters.

- Log-linear *guess and verify*. Guess that $\log p(x_t)$ is a linear function of the state variable:

  $$\log p(x_t) = A + Bx_t,$$

where we now try to find A and B:

$$\begin{aligned}\log p(x_t) &= \log E\left[e^{\log m(x_{t+1},x_t)+\log d(x_{t+1})+\log[p(x_{t+1})+1]} \mid x_t\right] \\ &= \log E\left[e^{\log \mu+x_t+\lambda\varepsilon_{t+1}+\delta_0+\delta_1 x_{t+1}+\log(A+Bx_{t+1}+1)} \mid x_t\right]\end{aligned}$$

The presence of the nonlinear function, $\log(A + Bx_{t+1} + 1)$ would appear to be a problem. We cannot verify our log-linear guess.

Campbell-Shiller approximation. Take a linear approximation of the function $\log[p(x_{t+1}) + 1] = \log[e^{\log p(x_{t+1})} + 1]$ around a constant value $\log \hat{p}$ so that it is approximately linear in $\log p(x_{t+1})$. The first-order Taylor series approximation around $\log \hat{p}$ is given by:

$$\begin{aligned}\log[p(x_{t+1})+1] &= \log[e^{\log p(x_{t+1})}+1] \\ &\approx \log(e^{\log \hat{p}}+1)+\left[\frac{e^{\log \hat{p}}}{e^{\log \hat{p}}+1}\right]\left(\log p(x_{t+1})-\log \hat{p}\right) \\ &= \gamma_0+\gamma_1 \log p(x_{t+1}),\end{aligned}$$

where

$$\gamma_1 = \frac{e^{\log \hat{p}}}{e^{\log \hat{p}}+1} \quad \text{and} \quad \gamma_0 = \log(e^{\log \hat{p}}+1)-\gamma_1 \log \hat{p}.$$

Given this approximation, we can use the same tricks for this equity valuation as we did for the multi-period bond calculation:

$$\begin{aligned}\log p(x_t) &= \log E\left[e^{\overbrace{\mu+x_t+\lambda\varepsilon_{t+1}}^{=\log m(x_{t+1},x_t)}+\overbrace{\delta_0+\delta_1 x_{t+1}}^{=\log d(x_{t+1})}+\overbrace{\gamma_0+\gamma_1(A+Bx_{t+1})}^{\approx\log[p(x_{t+1})+1]}} \mid x_t\right] \\ &= \mu+\delta_0+\gamma_0+\gamma_1 A+(1+\delta_1\varphi+\gamma_1 B\varphi)x_t+\log E\left[e^{(\lambda+\delta\sigma+\gamma_1 B\sigma)\varepsilon_{t+1}} \mid x_t\right] \\ &= \underbrace{\mu+\delta_0+\gamma_0+\gamma_1 A+(\lambda+\delta\sigma+\gamma_1 B\sigma)^2/2}_{=A}+\underbrace{(1+\delta_1\varphi+\gamma_1 B\varphi)}_{=B}x_t.\end{aligned}$$

We can solve for the log price-dividend ratio (at least up to the log-linear approximation by finding A and B that solve the two equations:

$$\begin{aligned}B &= 1+\delta_1\varphi+\gamma_1 B\varphi \\ \Rightarrow B &= (1+\delta_1\varphi)/(1-\gamma_1\varphi)\end{aligned}$$

$$\begin{aligned}A &= \mu+\delta_0+\gamma_0+\gamma_1 A+(\lambda+\delta\sigma+\gamma_1 B\sigma)^2/2 \\ \Rightarrow A &= (\mu+\delta_0+\gamma_0+(\lambda+\delta\sigma+\gamma_1 B\sigma)^2/2)/(1-\gamma_1).\end{aligned}$$

You may be wondering where we get the value for $\log \hat{p}$ around which to take the Taylor series approximation. Moreover, you may

also be wondering about the accuracy of that approximation. These are both excellent questions. The usual and customary procedure is to solve for the price-dividend ratio that is implied a model in which x_t is an *iid* random variable, x, with the same distribution as the long-run distribution of the stochastic process x_t:

$$x \sim \mathcal{N}\big(0,\ \sigma^2/(1-\varphi^2)\big).$$

As we saw with the Gordon growth model, this implies a constant value for the price-dividend ratio, $p(x_t) = p$, such that

$$\frac{p}{p+1} \quad = \quad E\big[e^{\log m(x_{t+1},x_t)+\log d(x_{t+1})}\big].$$

Define the variable in the exponent as $y_{t+1} = \log m(x_{t+1}, x_t) + \log d(x_{t+1})$, and note that this variable has an ARMA(1, 1) structure

$$\begin{aligned} y_{t+1} &= \log m(x_{t+1}, x_t) + \log d(x_{t+1}) \\ &= (\mu + \delta_0) + (1+\delta_1)x_t + \lambda w_{t+1} \\ \Rightarrow y_{t+1} - \varphi y_t &= (1-\varphi)(\mu+\delta_0) + \lambda w_{t+1} + [(1+\delta_1)\sigma - \lambda\varphi]w_t. \end{aligned}$$

Therefore, the long-run distribution for y_{t+1} is

$$y \sim \mathcal{N}\big(\mu+\delta_0,\ \lambda^2 + [(1+\delta_1)\sigma - \lambda\varphi]^2/(1-\varphi^2)\big).$$

This implies

$$\frac{p}{p+1} \quad = \quad e^{\mu+\delta_0+(1/2)\lambda^2+(1/2)[(1+\delta_1)\sigma-\lambda\varphi]^2/(1-\varphi^2)}.$$

Given the values of the parameters of the model $(\mu, \lambda, \varphi, \sigma, \delta_0, \delta_1)$, we can find the value of p that solves this equation, and that is the value we use to calculate γ_0 and γ_1 in the log-linear approximation.

How accurate is this approximation? Clearly, this will depend on the specifics of the model. For example, if the value of φ is close to 1 and the value of σ is large, then we would expect to see large and persistent departures of the conditional distribution for x_t from its long-run distribution, hence, the linear approximation around p might not be very accurate.

Markov-chain approximation. As an alternative to the Campbell-Shiller approximation, we could consider taking a discrete-state approximation to the AR(1) model. That is, imagine we chose a large but finite number of values for $x_t \in \{x_1, x_2, x_3, \ldots, x_N\}$. Assume that the values are evenly spaced so that $x_{j+1} - x_j = \epsilon$. We could then approximate the probability that $x_{t+1} = x_j$ conditional on $x_t = x_i$

using the standard normal probability distribution function, Φ as follows:

$$\begin{aligned}\pi_{ij} &= \text{Prob}[x_{t+1} = x_j | x_t = x_i] \\ &= \Phi\big([x_j + \epsilon/2 - \varphi x_i]/\sigma\big) - \Phi\big([x_j - \epsilon/2 - \varphi x_i]/\sigma\big),\end{aligned}$$

and for the end point simply

$$\begin{aligned}\pi_{i1} &= \text{Prob}[x_{t+1} \leq x_1 | x_t = x_i] \\ &= \Phi\big([x_1 + \epsilon/2 - \varphi x_i]/\sigma\big) \\ \pi_{iN} &= \text{Prob}[x_{t+1} \geq x_N | x_t = x_i] \\ &= 1 - \Phi\big([x_N - \epsilon/2 - \varphi x_i]/\sigma\big).\end{aligned}$$

There are many other ways we could approximate these probabilities, but this seems like the most direct. The grid of values we choose for x_t should be wide enough so that there isn't much probability that the stochastic process every gets to x_1 or x_N, *i.e.*, these should be extreme tail events of the unconditional distribution. In addition, the grid should be reasonably dense, *i.e.*, x_j should be reasonable close to x_{j+1}, so that discrete probabilities are close approximations to continuous probability densities.[3]

[3] Alternatively, we could start with a Markov chain as the primitive of the problem, rather than the continous AR(1). But as a practical matter, that model will have a lot of unknown parameters: N values for x_t, and $N^2 - N$ values of π_{ij}. Aside from simple examples like the $N = 2$ example we've seen previously, it won't be obvious how to choose values for all of these unknown parameters. And we know that the $N = 2$ example will not look very much like real-world data.

Note that once we've discretized the state variable, we will naturally have discrete values for the dividend process, $\log d_j = \delta_0 + \delta_1 x_j$, and the pricing kernel

$$\begin{aligned}\log m_{ij} &= m(x_{t+1} = x_j, x_t = x_i) \\ &= \mu + x_j + \lambda \underbrace{(x_j - \varphi x_i)/\sigma}_{=w_{t+1}} \\ &= \mu + (1 + \lambda/\sigma) x_j - (\lambda\varphi/\sigma) x_i.\end{aligned}$$

However we ultimately choose to the parameter values for the Markov chaine, once we have those We can now apply this discrete Markov chain to the equation for the price dividend ratio:

$$p_i = \sum_{j=1}^{N} \pi_{ij} \big[e^{\log m_{ij} + \log d_j}\big](p_j + 1), \quad i = 1, 2, \ldots, N.$$

Note that this is now a system of N linear equations in the N unknown values of the price-dividend ratio $\{p_1, p_2, \ldots, p_N\}$, which we can easily solve using standard methods.

Which approximation is "better", Campbell-Shiller or Markov chain? They each have their advantages. The log-normal structure of the Campbell-Shiller approximation means that we can easily extend

the solution method in a number of interesting directions just like we did with bond-pricing models. We could introduce additional stochastic processes as state variables to capture multiple sources or risk, we could model these state variables with more general ARMA dynamics, stochastic volatility, and Poisson disaster risk, all without deviating from the log-normal structure, hence the closed-form solutions for the parameters analogous to A and B in the simply AR(1) model. But how accurate will these approximations be? That's hard to tell. To know how close an approximation is the the exact solution, you pretty much have to know that exact solution. But the reason we adopted the approximation is because we don't know the exact solution. You can see the problem. In the Markov-chain approximation, we know we can make the solution very precise by increasing the size of the discrete grid, *i.e.*, making N large. The limitation in that approach, however, is that we quickly run out of computing capacity when we have multiple state variables or higher-order dynamics. Imagine we had a discrete grid of N points for each of 3 different state variables. In that case, the size of the Markov chain would be N^3 and we would need to solve N^3 equations for N^3 unknowns. Making N big to insure accuracy means making N^3 extremely big, which may not be feasible. As computing technology improves, we can solve bigger problems. But until quantum computing or some other radical innovation arrives, we will be constrained by the finite limitations of digital computers.

In other words, these different algorithms each comes with advantages and disadvantages. Which we choose will depend a lot on what particular questions we're trying to use the model to answer.

Pricing dividend strips.

If we started with our basic pricing equation

$$P(x_t) \quad = \quad E\big[m(x_{t+1}, x_t)[\underbrace{P(x_{t+1}) + D(x_{t+1})}_{\text{date-}t\text{ payoff}}] \mid x_t\big]$$

then repeatedly substituted for future prices and applied the law of iterated expectations, we would arrive at an equivalent representation for the equity price as the sum of expected discounted values of all future dividends

$$\begin{aligned} P(x_t) \quad = \quad & E\big[m(x_{t+1}, x_t)D(x_{t+1}) \mid x_t\big] + E\big[m(x_{t+1}, x_t)m(x_{t+2}, x_{t+1})D(x_{t+2}) \mid x_t\big] \\ & + E\big[m(x_{t+1}, x_t)m(x_{t+2}, x_{t+1})m(x_{t+3}, x_{t+2})D(x_{t+3}) \mid x_t\big] + \ldots \end{aligned}$$

We refer to each term in this infinite sum as a *dividend strip*. That is, each term can be thought of as the price at date t of an ownership

claim to a single dividend, rather than the entire stream of dividends. Denote the price of a dividend strip of the dividend that arrives at date $t+n$ as

$$\begin{aligned} Q_d^{(n)}(x_t) &= E\big[m(x_{t+1},x_t)m(x_{t+2},x_{t+1})\cdots m(x_{t+n},x_{t+n-2})D(x_{t+n}) \mid x_t\big] \\ &= \end{aligned}$$

which implies

$$P(x_t) = Q_d^{(1)}(x_t) + Q_d^{(2)}(x_t) + Q_d^{(3)}(x_t) + \cdots + Q_d^{(n)}(x_t) + \ldots$$

In other words, if we had a simple way to solve for $Q_d^{(n)}(x_t)$ then perhaps that might lead us to a simple way to solve for $P(x_t)$ as well.

The first step will be to transform the pricing equation so that it depends on dividend growth rates and price dividend ratios, then rewrite the equation as a recursion:

$$\begin{aligned} q_d^{(n)}(x_t) &= \frac{Q_d^{(n)}(x_t)}{D(x_t)} = E\big[m(x_{t+1},x_t)m(x_{t+2},x_{t+1})\cdots m(x_{t+n},x_{t+n-2})\frac{D(x_{t+n})}{D(x_t)} \mid x_t\big] \\ &= E\big[m(x_{t+1},x_t)m(x_{t+2},x_{t+1})\cdots m(x_{t+n},x_{t+n-2})d(x_{t+1})d(x_{t+2})\cdots d(x_{t+n}) \mid x_t\big] \\ &= E\Big[m(x_{t+1},x_t)d(x_{t+1})\underbrace{E\big[m(x_{t+2},x_{t+1})\cdots m(x_{t+n},x_{t+n-2})d(x_{t+2})\cdots d(x_{t+n}) \mid x_{t+1}\big]}_{=q_d^{(n-1)}(x_{t+1})} \Big| x_t\Big] \\ \Rightarrow q_d^{(n)}(x_t) &= E\big[m(x_{t+1},x_t)d(x_{t+1})q_d^{(n-1)}(x_{t+1}) \mid x_t\big] \\ \log q_d^{(n)}(x_t) &= \log E\big[e^{\log m(x_{t+1},x_t)+\log d(x_{t+1})+\log q_d^{(n-1)}(x_{t+1})} \mid x_t\big]. \end{aligned}$$

The form of this recursion should look very familiar from our models of multi-period bond pricing. In fact, we should be able to show that the same sort of long-linear *guess-and-verify* approach will work for these dividend strips just as easily as it did for bonds.

To see this, let's return to our AR(1) example. Recall the ingredients of that model:

$$\begin{aligned} x_t &= \varphi x_{t-1} + \sigma w_t \\ \log m_{t+1} &= \mu + x_t + \lambda w_{t+1} \\ \log d_t &= \delta_0 + \delta_1 x_t. \end{aligned}$$

We guess that $\log q_d^{(n)}(x_t)$ is a linear function of the state variable, with coefficients that can depend on n, the maturity of the dividend strip

$$\log q_d^{(n)}(x_t) = A_n + B_n x_t,$$

where we now use the recursion to find A_n and B_n to verify that the solution is indeed log-linear:

$$\begin{aligned}
\log q_d^{(n+1)}(x_t) &= \log E\left[e^{\log m(x_{t+1},x_t)+\log d(x_{t+1})+\log q_d^{(n-1)}(x_{t+1})} \mid x_t\right] \\
&= \log E\left[e^{\log \mu + x_t + \lambda\varepsilon_{t+1} + \delta_0 + \delta_1 x_{t+1} + A_n + B_n x_{t+1}} \mid x_t\right] \\
&= \mu + \delta_0 + A_n + (1+\delta_1\varphi + B_n\varphi)x_t + \log E\left[e^{(\lambda+\delta_1\sigma+B_n\sigma)\varepsilon_{t+1}} \mid x_t\right] \\
&= \underbrace{\mu + \delta_0 + A_n + (\lambda+\delta_1\sigma+B_n\sigma)^2/2}_{=A_{n+1}} + \underbrace{(1+\delta_1\varphi+B_n\varphi)}_{=B_{n+1}} x_t.
\end{aligned}$$

We therefore have verified the log-linear guess, and have derived recursive formulas for A_n and B_n:

$$\begin{aligned}
B_1 &= 1+\delta_1\varphi \\
B_n &= 1+\delta_1\varphi + B_{n-1}\varphi, \quad n = 2,3,\ldots \\
\Rightarrow B_n &= \left(\frac{1+\delta_1\varphi}{1-\varphi}\right)(1-\varphi^n)
\end{aligned}$$

$$\begin{aligned}
A_1 &= \mu + \delta_0 + (\lambda+\delta_1\sigma)^2/2 \\
A_n &= A_{n-1} + \mu + \delta_0 + (\lambda+\delta_1\sigma+B_{n-1}\sigma)^2/2, \quad n = 2,3,\ldots.
\end{aligned}$$

Again, this should seem very familiar given the obvious similarity to our models of multi-period bond pricing. We also know from that analysis that the log-normal recursive structure allows us to easily extend the model in a number of interesting directions: additional stochastic processes as state variables to capture multiple sources or risk, more general ARMA dynamics, stochastic volatility, and Poisson disaster risk, all without deviating from the log-linear structure and the recursive solutions. In other words, pricing dividend strips is no harder than pricing multi-period bonds.

But dividend strips are not really our end goal.[4] We are looking for the price of the entire sequence of dividends. Unfortunately, the log-linear structure doesn't extend, as we saw previously. However, it may provide a simple and useful approximation. Substitute the log-linear form into the equation for the price-dividend ratio

$$\begin{aligned}
p(x_t) &= \sum_{j=1}^{\infty} q_d^{(j)}(x_t) = \sum_{j=1}^{\infty} e^{\log q_d^{(j)}(x_t)} \\
&= \sum_{j=1}^{\infty} e^{\log q_d^{(j)}(x_t)} = \sum_{j=1}^{\infty} e^{A_j + B_j x_t}.
\end{aligned}$$

In general the infinite sum in this expression does not have a closed-form solution even when A_j and B_j are both known.

[4] Dividend strips are a type of derivative security that you may encounter in the real world, but it's a relatively small asset class. They can be replicated using options and futures, but only for short maturities. They are more commonly used simply an intermediate step to understanding equity prices.

To insure that $p(x_t)$ has a finite value, it is natural to assume that $\lim_{n\to\infty} Q_d^{(n)}(x_t) = 0$. That is, the value today of a payoff that arrives in the infinitely-distance future is zero. Since this also implies that $\lim_{n\to\infty} q_d^{(n)}(x_t) = 0$, a natural way to use the prices of dividend strips to approximate the price of an equity is to simply truncate the summation at a sufficiently large, but still finite, value

$$p(x_t) \quad = \quad \sum_{j=1}^{N} e^{A_j + B_j x_t} + \text{approximation error},$$

for N large enough that the approximation error is suitably small.

We need to be a little bit more careful with admissible value for the model's parameters. The assumption that $Q^{(n)}$ converges to zero as n gets large requires A_n to converge to $-\infty$ as n gets large. (Note that B_n converges to $(1+\delta_1)/(1-\varphi)$, so it is always finite with probability 1.) If we look more carefully at the solution for A_n we see that it is growing linearly in n, but if μ is a sufficiently large negative number, then the convergence is guaranteed. How large, of course, depends on all of the other value the model as dictated by the solution for A_n.[5] This is directly analogous to the $r > g$ requirement in the simple Gordon growth model that we started with.

[5] And the exact nature of that restriction will change as other details of the model change, *e.g.*, the stochastic process for the state variable.

CAPM

One of the most commonly used tools in equity valuation is the so-called *Capital Asset Pricing Model*, or simply CAPM. You've probably encountered this model in other finance classes, and may be wondering how this approach is related to our general pricing-kernel approach to asset pricing.

CAPM at its most general level states that the risk premium on any asset is proportional to the risk premium on a *market portfolio*. Without getting into too much detail about what this mysterious market portfolio might actually be, we can see how this feature of CAPM is a feature of any equilibrium asset pricing model. Recall our fundamental equation applied to *excess returns*, that is, returns in excess of the risk-free return, is

$$E_t\left[m_{t+1}(r_{t+1}^i - r_t^1)\right] = 0,$$

where r^i denotes the rate of return on any risky asset, and r^1 denotes the rate of return on a one-period risk-free bond. We can re-write this equation using the definition of a covariance to derive the risk premium

$$E_t\left[r_{t+1}^i - r_t^1\right] = -\frac{\mathrm{Cov}_t(m_{t+1}, r_{t+1}^i)}{E_t(m_{t+1})},$$

for any asset $i = 1, 2, \ldots, N$. Note that this implies that risk premiums are always proportional to one another. That is, consider the same equation for r^j_{t+1},

$$E_t\left[r^j_{t+1} - r^1_t\right] = -\frac{\mathrm{Cov}_t(m_{t+1}, r^j_{t+1})}{E_t(m_{t+1})},$$

which we can combine with the equation for r^i_{t+1} to get

$$E_t\left[r^i_{t+1} - r^1_t\right] = \left[\frac{\mathrm{Cov}_t(m_{t+1}, r^i_{t+1})}{\mathrm{Cov}_t(m_{t+1}, r^j_{t+1})}\right] E_t\left[r^j_{t+1} - r^1_t\right].$$

It therefore shouldn't come as a surprise to us that the CAPM predicts that the risk premium on any risky asset is proportional to the risk premium on the market portfolio, since *all* risk premiums exhibit that same proportionality.

CAPM: A theoretical version

We saw that when the representative agent has a multi-period expected utility model with $u(c) = \log(c)$, the pricing kernel is given by[6]

$$m_{t+1} = \beta c_t / c_{t+1}.$$

The goal of CAPM is to replace this theoretical consumption-based pricing kernel with something related to an asset-market outcome. The most likely reason for this switch is a lack of confidence in our ability to accurately measure c_t.

The return on wealth. How do we define wealth in this representative-agent economy? One way would be to use the literal meaning of wealth and value all of the assets that the agent has accumulated in the past and add to that the expected present value of all the income the agent will earn in the future. Alternatively, we could measure what that wealth my measuring what it will ultimately be used to purchase, since we know that there is always a binding budget constraint. It's often simpler to think about the value at date t of an ownership claim to all of the date-and-state-contingent consumption in the future. What was the point of saving in the past? What is the point of investing today? Earning income in the future? The obvious answer is that we want to fund our current and future consumption, since that is ultimately what is valued in the utility function. That is, wealth is only an intermediate step to consumption. In that case, wealth at date t, W_t, will be the expected discounted value of all future consumption

$$W_t = E_t[m_{t+1}c_{t+1}] + E_t[m_{t+1}m_{t+2}c_{t+2}] + E_t[m_{t+1}m_{t+2}m_{t+3}c_{t+3}] + \ldots$$

[6] You often here the log-utility special case described as having "income effects that perfectly offset substitution effects". We see that in action in this case. Think of increasing future income resulting in an increase in consumption by some amount, say x percent so that it becomes $(1+x)c_{t+j}$. Call this the income effect. Then ask how much extra date-t value does that extra consumption create? Note that the relative price that applies to date $t+j$ consumption is now $\beta^j c_t / [(1+x)c_{t+j}]$, so that this change in consumption also results in a corresponding reduction that relative price. In other words, anytime you make consumption bigger, you also make the marginal value of that consumption proportionately smaller, and the two effects perfectly offset each other.

When the pricing kernel is $m_{t+1} = \beta c_t / c_{t+1}$, *i.e.*, log expected utility, then this expression for wealth simplifies

$$\begin{aligned} W_t &= E_t[\underbrace{\beta(c_t/c_{t+1})}_{=m_{t+1}} c_{t+1}] + E_t[\underbrace{\beta^2(c_t/c_{t+2})}_{=m_{t+1}m_{t+2}} c_{t+2}] + E_t[\underbrace{\beta^3(c_t/c_{t+3})}_{=m_{t+1}m_{t+2}m_{t+3}} c_{t+3}] + \ldots \\ &= \beta c_t(1+\beta+\beta^2+\beta^3+\ldots) \\ &= \beta c_t/(1-\beta). \end{aligned}$$

What is the one-period return from owning this wealth asset with a date-t value of W_t? We will receive the equivalent of a dividend, c_{t+1}, and we will still own the value of the wealth asset going forward, W_{t+1}, that is

$$1 + r^W_{t+1} = \frac{c_{t+1} + W_{t+1}}{W_t} = \frac{c_{t+1} + \beta c_{t+1}/(1-\beta)}{\beta c_t/(1-\beta)} = \frac{c_{t+1}}{\beta c_t}.$$

In other words, the pricing kernel is the inverse of the return[7] on wealth,

$$m_{t+1} = 1/(1 + r^W_{t+1}).$$

Remember, this is not a general result, but rather an implication of using log expected utility.

[7] Note that we've switched from our customary notation and defined that return in terms of a *rate* of return, r^W, which is a more common form for CAPM models.

Since the inverse function isn't wildly nonlinear, it's common to simplify the model by using a linear approximation around, say, the mean $\bar{r}^W = E[r^W_{t+1}]$. Substitute that linear approximation,[8] $m_{t+1} \approx a_0 + a_1 r^W_{t+1}$, into the formula for the risk premium derived above to get

$$\begin{aligned} E_t\left[r^i_{t+1} - r^1_t\right] &\approx \left[\frac{\mathrm{Cov}_t(a_0 + a_1 r^W_{t+1}, r^i_{t+1})}{\mathrm{Cov}_t(a_0 + a_1 r^W_{t+1}, r^W_{t+1})}\right] E_t\left[r^W_{t+1} - r^1_t\right] \\ &= \left[\frac{\mathrm{Cov}_t(r^W_{t+1}, r^i_{t+1})}{\mathrm{Var}_t(r^W_{t+1})}\right] E_t\left[r^W_{t+1} - r^1_t\right] \\ &= b^i_t E_t\left[r^W_{t+1} - r^1_t\right], \end{aligned}$$

[8] A first-order Taylor series expansion would imply

$$1/(1 + r^W_{t+1}) \approx a_0 + a_1 r^W_{t+1},$$

where $a_1 = -1/(1+\bar{r}^W)^2$ and $a_0 = 1/(1+\bar{r}^W) - a_1 \bar{r}^W$.

where we have implicitly defined asset-i's "beta" as

$$b^i_t = \frac{\mathrm{Cov}_t(r^W_{t+1}, r^i_{t+1})}{\mathrm{Var}_t(r^W_{t+1})},$$

and I used the letter "b" even though it is commonly called "beta" since we already used the Greek letter β for the discounting parameters in the representative agent's utility function.

Conditional CAPM. Note that the model as written restricts the *conditional* risk premium and the "beta" depends on the conditional

covariance and the conditional variance. Deriving *conditional betas*, therefore, requires a specification of the dynamics of the state variables of the economy, *i.e.*, the information investors have that is useful in assessing risky future outcomes and how that information changes over time, along with a model of how asset returns, including the wealth return, depend on those state variables. Moreover, thinking of b_t^i as a parameter than can be estimated from data isn't feasible. Since it takes on a different value for each asset in each time period, we are guaranteed to have as many "parameters" as we have data points. As a result, this *conditional CAPM* isn't a very practical model.[9]

[9] Beware of *ad hoc* statistical procedures that claim to solve this dimensionality problem, *e.g.*, "rolling regressions". Ultimately, they do little more than wish this problem away and create a lot of spurious results along the way.

Unconditional CAPM. Does the model impose comparable restrictions for the expected value, *i.e.*, *unconditional* mean, of the risk premium, $E\left[r_{t+1}^i - r_t^1\right]$? That answer is yes, but we need to start back at the beginning:

$$\begin{aligned} E_t\left[m_{t+1}(r_{t+1}^i - r_t^1)\right] &= 0 \\ \Rightarrow E\left[E_t[m_{t+1}(r_{t+1}^i - r_t^1)]\right] &= E\left[m_{t+1}(r_{t+1}^i - r_t^1)\right] = 0. \end{aligned}$$

Note that this unconditional expected value restriction is a much weaker restriction to impose on asset returns. It doesn't restrict the expected value that is relevant for any and all information that the investor may have, but rather it just restricts the average *across* that information.

Now we repeat all of the same steps we used with the conditional model, but now using unconditional moments to arrive at the equation

$$\begin{aligned} E\left[r_{t+1}^i - r_t^1\right] &= \left[\frac{\mathrm{Cov}(r_{t+1}^W, r_{t+1}^i)}{\mathrm{Var}(r_{t+1}^W)}\right] E\left[r_{t+1}^W - r_t^1\right] \\ &= b^i E\left[r_{t+1}^W - r_t^1\right], \end{aligned}$$

where we have implicitly defined asset-i's "unconditional beta" as

$$b^i = \frac{\mathrm{Cov}(r_{t+1}^W, r_{t+1}^i)}{\mathrm{Var}(r_{t+1}^W)}.$$

This is the form of the model most commonly used when looked across a set of asset returns. The time series is used to estimate the unconditional means, variances, and covariances by averaging across time. This is now feasible because there is just one "beta" for each asset that applies in all time periods. Then the restriction implied by the unconditional CAPM can then be used to evaluate the relative magnitudes of average risk premiums *across* different assets.

Measuring the return on wealth. This theoretical CAPM won't be of much practical use unless we have a way of measuring r^W_{t+1}, *i.e.*, the return on wealth. The most common approach is to think about a broad portfolio of traded equity assets, something like the S&P 500 index. That seems like it ought to be pretty close to the value of all the capital in the economy, which ought to be pretty close to wealth. But what about things like *human capital* and the future labor income that it will generate that can also fund future consumption? Or other valuable, but non-traded assets like private partnerships? Or owner-occupied houses? Or a benevolent government that provides valuable public goods? And what do we do about foreign ownership of assets?

When combined with the hard-to-justify restriction of log expected utility (What if the representative consumer is more risk averse than that?), with the many measurement issues associated with r^W_{t+1}, the usefulness of this model would seem to be severely limited. As a result, that's not typically how applied financial economists think about the CAPM. Instead, they approach the problem from a statistical *fitting* perspective that we turn to now.

CAPM: A statistical version

What is attractive about the theoretical version of CAPM is its ability to characterize risk premiums without the need to measure consumption. What is unattractive about it still requires a measurement of the return on *wealth*, which may be just as challenging as measuring consumption itself. An alternative way to think about CAPM, therefore, is simply as the specification of an *approximate* pricing kernel.

Imagine that we have a broad-based *well-diversified* portfolio of assets with a return, r^P_{t+1}, that is relatively easy to measure.[10] Note that we are *not* assuming that this is the return to wealth. It could be, of course, but that's not necessary for what follows. We know that we can always decompose the pricing kernel into two pieces: one that depends linearly on r^P_{t+1}, and the other piece is everything else, ε_{t+1}, such that

$$m_{t+1} = \gamma_0 + \gamma_1 r^P_{t+1} + \varepsilon_{t+1}.$$

This approach is motivated by the theoretical CAPM above, but with an explicit recognition that the portfolio return that, r^P_{t+1}, is serving only as an approximation of the pricing kernel rather than an explicit measurement of the return to wealth.

How do we choose the parameters of this approximate model, γ_0 and γ_1? We would like the linear function of r^P_{t+1} to be *as close as possible*

[10] This is a tricky assumption since the concept of *diversification* quickly becomes circular. That is, a *well-diversified* portfolio implicitly requires a definition of the riskiness of the assets in that portfolio, since it's that risk that is being *diversified*. How do we assess the riskiness of an asset if we don't have a model of the pricing kernel?

to the true pricing kernel, m_{t+1}. In the context of random variables, a natural way to do this is to make the variance of their difference as small as possible, *i.e.*, minimize the variance of ε_{t+1}, which is given by

$$\mathrm{Var}(m_{t+1} - \gamma_0 - \gamma_1 r^P_{t+1}) = \mathrm{Var}(m_{t+1}) + \gamma_1^2 \,\mathrm{Var}(r^P_{t+1}) - 2\gamma_1 \mathrm{Cov}(m_{t+1}, r^P_{t+1}).$$

Choose γ_1 to minimize that variance

$$\min_{\gamma_1} \;\; \mathrm{Var}(m_{t+1} - \gamma_0 - \gamma_1 r^P_{t+1}) \;\; \Rightarrow \;\; \gamma_1 = \frac{\mathrm{Cov}(m_{t+1}, r^P_{t+1})}{\mathrm{Var}(r^P_{t+1})}.$$

It would also seem natural if the linear function of r^P_{t+1} was equal to the pricing kernel, m_{t+1}, on average, *i.e.*, $E[\varepsilon_{t+1}] = 0$, which implies

$$\gamma_0 = E(m_{t+1}) - \gamma_1 E(r^P_{t+1}).$$

This process for choosing γ_0 and γ_1 should remind you of a *linear regression* in statistics. We can't actually use a regression to identify γ_0 and γ_1, of course, since we don't know m_{t+1} – if we did we would just use it – so this regression interpretation helpful only at a conceptual, not a practical level.

Note that choosing the parameters this way guarantees that the approximate pricing kernel. $\gamma_0 + \gamma_1 r^P_{t+1}$, works as well as the actual pricing kernel in the equations for the unconditional equity premium for the portfolio return, $E[r^P_{t+1} - r^1_t]$:

$$\begin{aligned} E\left[r^P_{t+1} - r^1_t\right] &= -\frac{\mathrm{Cov}(\gamma_0 + \gamma_1 r^P_{t+1}, r^P_{t+1})}{E(\gamma_0 + \gamma_1 r^P_{t+1})} \\ &= -\frac{\frac{\mathrm{Cov}(m_{t+1}, r^P_{t+1})}{\mathrm{Var}(r^P_{t+1})} \overbrace{\mathrm{Cov}(r^P_{t+1}, r^P_{t+1})}^{\mathrm{Var}(r^P_{t+1})}}{E(m_{t+1})} = -\frac{\mathrm{Cov}(m_{t+1}, r^P_{t+1})}{E(m_{t+1})}. \end{aligned}$$

What we really care about, however, is how well the approximate pricing kernel works for any arbitrary asset, r^i_{t+1}. Recall the equation that showed the proportionality of risk premiums. Apply this to r^i_{t+1} and r^P_{t+1}:

$$\begin{aligned} E\left[r^i_{t+1} - r^1_t\right] &= \left[\frac{\mathrm{Cov}(m_{t+1}, r^i_{t+1})}{\mathrm{Cov}(m_{t+1}, r^P_{t+1})}\right] E\left[r^P_{t+1} - r^1_t\right] \\ &= \left[\frac{\mathrm{Cov}(\gamma_0 + \gamma_1 r^P_{t+1} + \varepsilon_{t+1}, r^i_{t+1})}{\mathrm{Cov}(\gamma_0 + \gamma_1 r^P_{t+1} + \varepsilon_{t+1}, r^P_{t+1})}\right] E\left[r^P_{t+1} - r^1_t\right] \\ &= \left[\frac{\mathrm{Cov}(r^P_{t+1}, r^i_{t+1})}{\mathrm{Var}(r^P_{t+1})} + \frac{\mathrm{Cov}(\varepsilon_{t+1}, r^i_{t+1})}{\mathrm{Cov}(m_{t+1}, r^P_{t+1})}\right] E\left[r^P_{t+1} - r^1_t\right]. \end{aligned}$$

The first term in the large brackets is what we're looking for: a "beta" for asset-i relative to the portfolio return, r^P_{t+1}. But there's nothing in what we've assumed so far that says that the second term will be zero. In other words, we need to make an additional *assumption* that

$$\mathrm{Cov}(\varepsilon_{t+1}, r^i_{t+1}) = 0.$$

In essence, this says that the only part of the variation in r^i_{t+1} that is related to the pricing kernel, m_{t+1}, is *completely* captured by the r^P_{t+1}. It is extremely important to remember that this assumption is *not* motivated by any economics. It is an arbitrary mechanical assumption that is made solely for the purpose of rationalizing this particular portfolio, r^P_{t+1}, as a substitute for the unknown pricing kernel, m_{t+1}. Moreover, without specifying a an explicit model for m_{t+1}, there's no way to assess this assumption.

If we're willing to make that additional leap, then we have a version of CAPM

$$E\left[r^i_{t+1} - r^1_t\right] \quad = \quad b^i E\left[r^P_{t+1} - r^1_t\right],$$

where the "beta" is defined as

$$b^i = \frac{\mathrm{Cov}(r^P_{t+1}, r^i_{t+1})}{\mathrm{Var}(r^P_{t+1})}.$$

The CAPM "alpha". What would we conclude if we estimated the "beta" for an asset using observed data but found that the average risk premium on that asset was different than the one predicted by this statistical version of CAPM? That is, what if the data looked like they came for the model

$$E\left[r^i_{t+1} - r^1_t\right] \quad = \quad a^i + b^i E\left[r^P_{t+1} - r^1_t\right],$$

where $a^i \neq 0$. The intercept term in this equation is often referred to as "alpha" (we use the letter "a" here to avoid confusion with our previous use of α to denote a preference parameter).

One popular conclusion is that any asset with $a^i \neq 0$ presents a profitable investment opportunity. For example, if $a^i > 0$, then an investment in asset i pays a return that is bigger than is needed to compensate the investor for the riskiness of that asset. This isn't an arbitrage opportunity, but it is a *good deal*. Let's form a hedge fund, develop a profitable trading strategy based on assets with $a^i \neq 0$, and get really rich! Wait a second. Don't forget that we made an arbitrary mechanical assumption that $\mathrm{Cov}(\varepsilon_{t+1}, r^i_{t+1}) = 0$. We had

no theoretical or empirical basis for this assumption. What would we expect to see if that assumption is wrong? We would see $a^i \neq 0$. In other words, we can't ever distinguish between two possibilities: assets are mis-priced by the market hence they have extraordinary returns, or our statistical CAPM model is simply wrong.[11]

[11] It shouldn't come as a surprise, therefore, that most hedge funds quickly go bust. The half life of a hedge fund is estimated to be somewhere between twelve and eighteen months.

The CAPM regression. If the CAPM equation held exactly in data, then estimating the "beta" would be a trivial exercise. It's just the ratio of the average risk premium on asset i and the average risk premium on the broad-based portfolio that is standing in for the pricing kernel, so replacing expected values with sample averages seems like a straightforward way to estimate "beta". This is tautological. What if we wanted to use real-world data not just to define an asset's "beta" but also to evaluate how well the data conform to the predictions of the model? Traditionally, this is done by using the basic CAPM equation to define a linear regression, which then forms the basis for statistical estimation and hypothesis testing.

Rewrite the basic CAPM equation by adding and subtracting the random variables themselves which leaves the equation unchanged

$$E\left[r^i_{t+1} - r^1\right] \quad = \quad b^i E\left[r^P_{t+1} - r^1\right] + \underbrace{(r^i_{t+1} - r^1) - (r^i_{t+1} - r^1)}_{=0} + \underbrace{b^i(r^P_{t+1} - r^1) - b^i(r^P_{t+1} - r^1)}_{=0}.$$

Notice that we've also dropped the time subscript from the risk-free interest rate. In other words, we're now assuming that the risk-free rate isn't just known at date t, it is literally constant for all t. Now rearrange the terms in this expression to get

$$r^i_{t+1} - r^1 \quad = \quad b^i(r^P_{t+1} - r^1) + e^i_{t+1},$$

where we have defined an *error term*, e^i_{t+1}, such that

$$e^i_{t+1} = (r^i_{t+1} - r^1) - E\left[r^i_{t+1} - r^1\right] + b^i(r^P_{t+1} - r^1) - b^i E\left[r^P_{t+1} - r^1_t\right] - E\left[r^i_{t+1} - r^1_t\right].$$

This error term has two important features: It has an expected value of zero, $E[e^i_{t+1}] = 0$, and it is uncorrelated with r^P_{t+1},

$$\begin{aligned}
\mathrm{Cov}(r^P_{t+1}, e^i_{t+1}) &= \mathrm{Cov}(r^P_{t+1}, r^i_{t+1} - b^i r^P_{t+1}) \\
&= \mathrm{Cov}(r^P_{t+1}, r^i_{t+1}) - b^i \mathrm{Cov}(r^P_{t+1}, r^P_{t+1}) \\
&= \mathrm{Cov}(r^P_{t+1}, r^i_{t+1}) - \left[\frac{\mathrm{Cov}(r^P_{t+1}, r^i_{t+1})}{\mathrm{Var}(r^P_{t+1})}\right] \mathrm{Var}(r^P_{t+1}) \\
&= 0.
\end{aligned}$$

In other words, we have used the restrictions of CAPM (and the assumption of a constant interest rate) to create a simple linear regression model,

$$r^i_{t+1} - r^1 \quad = \quad a^i + b^i(r^P_{t+1} - r^1) + e^i_{t+1},$$

that can be estimated in the customary fashion using *ordinary least squares*. This provides both an estimate of b^i and a standard *test* of the model, $a^i = 0$.

End of the story? Not even close. Estimation of this regression model with a wide variety of different assets and different time periods consistently produces results that contradict the basic predictions of the model. Not only is a^i frequently found to be different than zero for many assets, the error term e^i_{t+1} typically looks nothing like a standard *regression residual*. Rather it looks like a stochastic process that has a lot of persistent dynamics. In other words, we seem to have information at t that is useful in predicting movements in returns at $t+1$ that have nothing to do with CAPM's measurement of risk. Should we conclude that there is simply a lot of free money out there just waiting for one of us to come along and pick it up? Almost certainly not! It is far more likely that r^P_{t+1} isn't a very good approximation for the true pricing kernel. Moreover, the dynamics evident in e^i_{t+1} suggest that ignoring the dynamic implications of CAPM by looking only at unconditional moments was not a very productive approach. The dynamics end up mattering a lot!

Because of these numerous shortcomings of CAPM – both the theoretical and the statistical versions – a number of other statistical approaches to approximating the pricing kernel have been attempted. They generally take the form of so-called *factor models*. These differ substantially both in motivation and in application from the factor models we studied in the context of multi-period bond pricing, so we will discuss them separately next.

Factor Models

On response to the failure of the CAPM to capture the behavior or real-world equity returns is to simply keep expanding the number of terms in the approximation of the pricing kernel. That is, rather than simply $m_{t+1} = \gamma_0 + \gamma_1 r^P_{t+1} + \varepsilon_{t+1}$, we could assume some larger number of sources of risk, often called *factors*,

$$m_{t+1} = \gamma_0 + \gamma_1 F_{1t+1} + \gamma_2 F_{2t+1} + \cdots + \gamma_N F_{Nt+1} + \varepsilon_{t+1},$$

and maintain the arbitrary and mechanical assumption that

$$\mathrm{Cov}(\varepsilon_{t+1}, r^i_{t+1}) = 0.$$

This seems like it ought to be a step forward, but in reality, we need to place substantially more structure on these so-called risk factors to make any progress.[12] Without the structure of an economic theory,

[12] The *factor* terminology derives from the similarity of this setup with a statistical method called *factor analysis*.

we won't know how to measure these factors, or even how many of them we need. For example, is this an N-factor model, or a 1-factor model? Since we don't know what these factors are or how they behave, why don't we define a factor called $\hat{F}_{t+1} = F_{1t+1} + (\gamma_2/\gamma_1)F_{2t+1} + \cdots + (\gamma_N/\gamma_1)F_{Nt+1}$ so that we have a 1-factor model of the pricing kernel

$$m_{t+1} = \gamma_0 + \gamma_1 \hat{F}_{1t+1}.$$

This is analogous to saying that there are N state variables that affect the return to wealth and the return to wealth determines the pricing kernel. We could use those N factors to approximate the pricing kernel, or we could just simply use 1 factor, the return to wealth. Asset returns would be the same in either case. The question becomes one of feasible measurement rather than dimension. Eugene Fama has referred to this idea as a *fishing license*. You can just fish around for factors in a totally undisciplined way and when you're done, you can justify your results by claiming that you've simply identified the variables that drive the difficult-to-measure return to wealth or the difficult-to-measure consumption growth rate. Clearly, we need some way to discipline this statistical procedure so that we don't just over-fit a particular sample and claim that spurious correlations are in fact economic structure.

In addition, if we are going to make the model more complicated in an attempt to make it look more realistic, we should do something to take account of the observed dynamic behavior of returns at the same time.

Information-managed portfolio returns. Let's go back to our basic asset-pricing equation

$$P^i(z_t) = E\left[m_{t+1}X^i_{t+1} \,|\, z_t\right], \quad i = 1, 2, \ldots, n$$

where now the information available at time t is explicit in the expectation conditional on date-t state variable, z_t, the date-$t+1$ payoff (*i.e.*, resale price and dividend) is denoted X^i_{t+1}, and there are a finite number of assets, n, in the economy. Consider an arbitrary function of the state variable, $g(z_t)$, then we also know that by the law of iterated expectations, this equation also implies

$$E\left[g(z_t)P^i(z_t)\right] = E\left[m_{t+1}g(z_t)X^i_{t+1}\right],$$

which we could write equivalently as

$$E\left[m_{t+1}R^g_{t+1}\right] = 1,$$

where we've implicitly defined the return on this function is $R^g_{t+1} = g(z_t)X^i_{t+1}/E[g(z_t)P^i(z_t)]$. You can interpret $g(z_t)$ as an *information-based trading stategy* that says "whenever the state of the economy, z_t, takes on a particular value, I will trade $g(z_t)$ of asset i". R^g_{t+1} is the return from this strategy.

There's a theorem in stochastic processes that implies – loosely speaking – that if the unconditional asset-pricing restriction in the last equation holds for *every* information-managed trading strategy involving the n primitive assets, then the *conditional* asset-pricing condition holds as well. That is, *if for all* $g(z_t)$,

$$E\big[g(z_t)P^i(z_t)\big] = E\big[m_{t+1}g(z_t)X^i_{t+1}\big],$$

then it necessarily follow that

$$P^i(z_t) = E\big[m_{t+1}X^i_{t+1} \,|\, z_t\big], \quad i = 1, 2, \ldots, n.$$

What this means from a practical perspective is that if we have an unconditional version of the CAPM that is in some sense *general enough* (remember the theorem requires *all* functions of z_t), then implicitly, we have captured all of the dynamics implied by the conditional model.

But how do we operationalize *general enough*? Here's what is typically done. Use the date-t information to form a portfolio of the n primitive assets, $\omega^i(z_t)$, $i = 1, 2, \ldots, n$. Then calculate the rate of return on this *information managed portfolio* as

$$r^\omega_{t+1} = \frac{\sum_{i=1}^n \omega^i(z_t)X^i_{t+1}}{\sum_{i=1}^n \omega^i(z_t)P^i(z_t)} - 1.$$

Use this portfolio to form an approximate *statistical* pricing kernel that is linear in the managed portfolio return, $m_{t+1} = \gamma_0 + \gamma_\omega r^\omega_{t+1} + \varepsilon_{t+1}$, and use that as before to motivate the regression model

$$r^i_{t+1} \quad = \quad a^i + b^i_\omega r^\omega_{t+1} + e^i_{t+1}.$$

If the data suggest that $a^i \neq 0$ and e^i_{t+1} is not *iid*, then we haven't exhausted the value of the date-t information, so we look for an additional *information-managed portfolio* to add to the pricing kernel approximation. Let's say that this new managed portfolio is denoted $\eta^i(z_t)$, $i = 1, 2, \ldots, n$, with a rate of return

$$r^\eta_{t+1} = \frac{\sum_{i=1}^n \eta^i(z_t)X^i_{t+1}}{\sum_{i=1}^n \eta^i(z_t)P^i(z_t)} - 1.$$

We then extend our pricing kernel approximation to $m_{t+1} = \gamma_0 + \gamma_\omega r^\omega_{t+1} + \gamma_\eta r^\eta_{t+1} + \varepsilon_{t+1}$ and use that as before to motivate the regression model

$$r^i_{t+1} \quad = \quad a^i + b^i_\omega r^\omega_{t+1} + b^i_\eta r^\eta_{t+1} + e^i_{t+1},$$

and repeat.

The belief is that not literally *every* information-based trading strategy contains useful asset-pricing information, but rather just a relatively small but *carefully chosen* set of such portfolios. How small? Some people are content with three, others believe it's four, and there's even a growing belief that you need five of these information-managed portfolios as factors to capture all of the features of the data.

And what does "carefully chosen" mean? That's where the wheels tend to come off the wagon. Generally, people have loosely formulated and casually stated "theories" of what kinds of risks are being valued by financial markets. Here's a partial list of the factors that you're most likely to come across: (1) the "market portfolio" which is usually a value-weighted return on a broad set of traded equity assets (note that the value weights change from period to period, hence, are a simple example of an information-managed portfolio); (2) a "size-based portfolio" where you form a portfolio by selling big firms and buying small firms (note that whether a firm is big or small is something that changes over time which is another example of an information-based strategy); (3) "book to market" in which you buy firms that have a high accounting value relative to its current market price and sell the ones that have high market value relative to their accounting value; (4) "mean reversion" is a portfolio in which you sell the firms whose returns are high relative to their forecast and buy firms with returns below their forecast; (5) "momentum" is when you do the opposite of mean reversion; (6) "profitability" buys the firms with high operating profits and sells the firms with low operating profits; and (7) "investment" buys firms that invest conservatively and sells firms that invest aggressively. And then there are portfolios that combine these managed portfolios, *e.g.*, buy firms that are both small and have high book-to-market values and sell firms that are both large and have low book-to-market values, *etc*.

Yes, this ends up not being terribly well-disciplined relative to our theoretical models. Is this approach really using the dynamic properties of returns or is it just another type of the fishing license that Fama dislikes? For most people, it's hard to see much difference.

Conclusion

Equity pricing models should build on the same theory as all of our pricing-kernel models. There is a huge demand for practical research in this area, however, which has motivated a tremendous volume of statistical models that are neither firmly grounded in economic theory, or ultimately very useful for asset pricing. In the end, there is no easy way around the need to explicitly model the pricing kernel whether we're talking about equity assets, multi-period bonds, or their derivatives.

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